

SKILLS

- **AI / ML:** Python, PyTorch, Hugging Face, LLM training and evaluation, inference, RAG, agentic workflows, tool use
- **Distributed Training:** PyTorch FSDP/DDP, DeepSpeed ZeRO, mixed precision, gradient checkpointing, NCCL, GPU profiling
- **Systems Engineering:** TypeScript, Go, SQL, distributed systems, REST APIs, Kafka, PostgreSQL, Redis, Elasticsearch
- **Compute & Operations:** Kubernetes, Docker, AWS, GCP, Terraform, CI/CD, OpenTelemetry, SLOs, incident response
- **Secure AI:** Sovereign AI, Responsible AI, model governance, IAM, Zero Trust, data residency, auditability

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Applied AI *Nov 2022 – Present*
 - **Agentic AI Systems:** Building ASB Assist, an LLM + RAG system that converts natural-language requirements into compliant architecture diagrams and service recommendations; implementing Python orchestration, retrieval, tool use, evaluations, grounding controls, guardrails, and audit trails.
 - **Sovereign AI & Model Governance:** Operationalized an enterprise AI governance framework spanning model risk, Responsible AI, IAM, data residency, regulatory controls, and auditable deployment across AWS, GCP, and air-gapped OpenShift environments.
 - **Production Software Engineering:** Designed and shipped a patent-pending TypeScript/React platform that maps 1,000+ architecture patterns to real-time service selections and automated certification; authored 99.6% of production code and all core recommendation logic.
 - **Distributed Data Infrastructure:** Architected an OpenTelemetry platform using Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB/day across 10,000+ systems, with resilient pipelines, schema contracts, observability, and vendor-independent operations.
 - **Technical Leadership:** Lead 5 engineers through research spikes, system design, implementation, code review, testing, ADRs, and operational handoff while aligning product, security, risk, and infrastructure stakeholders.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer *Aug 2021 – Jul 2022*
 - **Production ML & Platform Delivery:** Deployed transformer-based document classification behind monitored inference APIs and built the surrounding React/TypeScript, Node.js, Hyperledger Fabric, and AWS EKS platform with auditable, HIPAA-aligned workflows.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op *Apr 2022 – Nov 2022*
 - **Public-Interest Software:** Modernized a federal tax platform from Apache Struts to React/TypeScript and contributed to APIs serving millions of Canadians, with WCAG accessibility and automated quality gates.

SELECTED RESEARCH & PROJECTS

- **Distributed LLM Training Lab:** Built a PyTorch/Hugging Face training harness for multi-GPU fine-tuning with FSDP/DDP, DeepSpeed ZeRO, mixed precision, gradient checkpointing, checkpoint recovery, experiment tracking, and reproducible evaluation.
- **Enterprise Agent Evaluation Harness:** Built offline task suites for a Python RAG agent, measuring groundedness, tool-selection accuracy, citation validity, latency, and regression risk with end-to-end traces.
- **DCS-BBN Decentralized Storage:** Designed a privacy-preserving peer-to-peer storage network using Solidity, Hyperledger Fabric, and zk-SNARKs; published at Springer's International Conference on Mobile Computing.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing *Jan 2021 – Aug 2022*
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) *Completed 2024*
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 *Aug 2016 – Aug 2020*